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Visualizing United States Greenhouse Gas Emissions

Abstract

Greenhouse gas (GHG) emissions are a central topic in climate change and global warming discussions; however, many people are unaware of the magnitude of GHG emissions and cannot conceptualize such large numbers. This project aims to visualize the GHG's in our environment in a way that can be easily interpreted by audiences without scientific backgrounds.

Overview

Using a Tableau Storyboard, each slide will provide different visualizations of user selected GHG's such as carbon, methane, and nitrous oxides. Total carbon equivalence (CO₂e) will also be an option.

The first two slides will be general comparative maps for total emissions based on state/county using user-defined inputs (parameters). The first slide will visualize user selected GHG country wide, comparing state total emissions. The user will also choose the year and whether to display this map with graduated colors or as a cartogram/size display. The second slide will be similar but narrow the focus to a user-selected state that compares counties emissions using the same, user-defined symbology as before. Numeric data, such as tables and graphs will be incorporated on these slides or in proceeding slides that connect to these slides. A third slide could be added to show change over time as well.

The proceeding 3 slides will focus on unique ways to represent emissions data with one slide dedicated to each GHG. User inputs will be year, display, and a time slider. An example is using a visual representation of how many pounds of coal would be burned to reach the CO₂ emissions in a given year, where the amount of carbon is represented by an increasing pile of coal. Another example is the amount of US land that would need to become forest to sequester a given amount of carbon in one year. How many trips along Route 66 based on different cars that I would pre-define (sedan, van, truck, etc.) could be another example. Cows for methane, and fertilizer/agriculture for nitrous oxides. This part of the project would involve my own calculations and dimensional analysis. The math and sources would be provided as well.

Technology

Tableau Public, Excel

Data

EPA emissions data
EPA fuel economy data
Peer reviewed sources for numbers.

Methodology

- Download data
- Spatially join data
- Create parameters and filters
- Create calculated fields
- Create tables and graphs
- Create dashboards

Inspiration

In a previous GIS course, I completed an ArcGIS Pro project on CO₂e in Iowa and thought to expand that idea to the contiguous United States using Tableau. As my data wrangling and spatial data knowledge have improved over the years, expanding on this previous project by restructuring it will show the progress I've made.

Potential Challenges

- Finding reliable data
- Calculations
- Finding a way to visualize things correctly

Timeline for completion

- Data gathered and cleaned by April 12th
- Tableau outlines for all the visualizations, start calculations by April 19th
- Begin integrating calculations and parameters by April 26th
- Finish calculations and parameters by May 3rd
- Composition, tidying, and completion by May 10th (at the latest)

AI was not used in any form to write this proposal.